

AKIAK, AK, USA

WMO: 702686

Lat: 60.903N

Lon: 161.231W

Elev: 30

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00106

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-18.0	-29.4	1.1	-22.5	-24.3	1.4	-17.6	41.2	8.2	37.1	9.7	5.9	30	0.926

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	65.0	57.8	62.2	56.0	59.3	55.1	59.0	63.1	57.2	60.7	55.6	58.6	11.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.1	69.5	60.5	55.4	65.3	58.7	54.0	62.2	57.4	25.7	63.1	24.5	60.6	23.6	58.6	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.9	31.5	27.6	-23.5	73.5	5.1	5.9	-27.2	77.7	-30.2	81.2	-33.1	84.5	-36.8	88.7
			-22.5	62.3	6.2	4.3	-27.0	65.4	-30.6	67.9	-34.1	70.4	-38.7	73.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.5	8.0	12.9	11.2	23.8	37.7	48.0	52.2	52.7	47.7	34.0	21.8	15.2	
	DBStd	19.47	16.99	14.92	13.10	10.52	7.93	5.67	5.15	3.83	5.03	10.03	11.96	13.42	
	HDD50	7380	1301	1037	1204	786	386	104	33	15	94	497	845	1078	
	HDD65	12584	1766	1457	1669	1236	846	511	400	381	518	962	1295	1543	
	CDD50	273	0	0	0	0	5	43	100	99	26	0	0	0	
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0	
	CDH74	14	0	0	0	0	0	0	14	0	0	0	0	0	
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0	
Wind	WSAvg	13.0	12.9	14.6	14.9	13.3	11.4	11.8	11.5	11.8	12.5	13.4	14.1	13.6	
Precipitation	PrecAvg	19.2	0.9	0.8	0.7	0.7	1.1	1.7	2.5	3.2	3.0	1.7	1.7	1.1	
	PrecMax	23.6	2.6	3.0	3.6	2.1	2.7	4.3	3.9	6.3	6.9	4.3	4.1	3.4	
	PrecMin	12.5	0.1	0.0	0.1	0.0	0.1	0.3	0.7	1.1	1.0	0.2	0.0	0.2	
	PrecStd	2.9	0.6	0.6	0.7	0.5	0.7	0.7	0.9	1.1	1.4	1.0	1.0	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	35.4	36.0	47.4	61.2	67.3	73.8	66.1	61.2	51.8	41.9	38.4	
		MCWB	34.7	33.7	34.5	41.1	51.9	58.2	63.1	58.2	54.7	49.3	40.3	36.8	
	2%	DB	34.6	34.4	34.2	42.3	55.5	63.0	66.4	63.3	57.4	49.2	40.2	36.4	
		MCWB	33.3	33.2	33.2	38.8	49.5	55.5	60.1	56.1	53.5	47.2	39.0	35.0	
	5%	DB	33.5	33.7	32.7	38.1	52.3	59.4	62.8	60.8	55.4	47.2	38.2	34.4	
		MCWB	32.7	32.9	32.0	36.0	47.5	53.9	57.2	55.8	52.4	45.6	37.0	33.4	
	10%	DB	31.6	32.3	30.3	35.6	48.1	56.2	59.2	58.3	54.2	45.3	36.5	33.5	
		MCWB	31.1	31.7	29.9	33.9	45.0	52.0	55.5	55.2	51.4	43.7	35.3	32.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	34.4	34.9	42.3	53.1	59.0	66.1	59.9	57.1	50.2	41.0	37.4
			MCDB	36.2	34.7	35.5	46.9	58.5	65.7	72.7	63.9	59.6	51.6	41.4	37.7
2%		WB	33.7	33.4	33.4	38.9	50.1	56.6	60.5	58.1	54.6	47.8	39.0	35.0	
		MCDB	34.4	33.8	33.9	42.0	55.5	61.6	64.7	61.0	56.6	49.0	39.9	35.9	
5%		WB	32.5	32.8	32.0	36.1	47.6	54.4	57.9	56.7	53.0	46.0	37.2	33.6	
		MCDB	33.2	33.3	32.6	38.1	51.5	58.3	61.8	59.2	55.0	47.1	38.0	34.3	
10%		WB	30.8	31.7	30.1	34.0	45.6	52.4	55.8	55.4	51.5	44.0	35.5	32.3	
		MCDB	31.4	32.2	30.6	35.2	48.4	56.0	59.1	58.0	53.4	45.3	36.4	33.1	
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.8	12.4	11.3	9.7	8.6	7.1	7.2	7.1	7.7	9.0	10.3	
		MCDBR	9.8	7.3	9.8	10.9	15.9	15.1	14.0	11.2	9.3	6.2	6.5	7.7	
		MCWBR	9.5	7.1	9.3	8.2	10.2	9.1	7.8	6.0	6.0	5.5	6.7	7.3	
	5% WB	MCDBR	9.7	7.9	9.3	10.5	15.1	13.4	12.4	9.0	7.5	5.5	6.5	8.1	
		MCWBR	9.4	7.8	9.0	8.0	10.0	8.6	7.4	5.6	5.6	5.4	6.7	7.8	
Clear-Sky Solar Irradiance	taub	0.228	0.283	0.290	0.349	0.380	0.366	0.370	0.382	0.342	0.296	0.247	0.215		
	taud	2.084	2.172	2.234	2.181	2.280	2.393	2.421	2.374	2.447	2.415	2.076	1.926		
	Ebn at Noon	164	220	265	266	267	273	269	255	246	220	147	104		
	Edh at Noon	12	22	29	37	37	34	32	31	24	17	12	9		
All-Sky Solar Radiation	RadAvg	116	338	926	1431	1612	1657	1369	1067	752	381	136	58		
	RadStd	18	43	100	126	154	135	132	117	88	30	21	8		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (7 neighbors)		N/A	+3.01	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page